Ahmed Fathy

Software Engineer
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Education

- **Cairo University, Faculty of Engineering** — Giza, Egypt
  B.Sc. in Computer Engineering; GPA: 3.6 — *2022*

Experience

- **Al Nada Scientific Office** | *Technical Support Trainee* — *Cairo, Egypt | June 2022 – 2024*
  - **Diagnostics & Maintenance**: Troubleshot workstations and network equipment; performed firmware updates and preventive maintenance.
  - **Documentation**: Created systematic troubleshooting documentation to streamline hardware issue resolution.

Student Activities

- **IEEE** | *UI/UX Designer - Frontend Team Leader* — *Cairo | Oct 2025 – Present*
  - **Portfolio Platform Development**: Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
- **Technical Center for Career Development (TCCD)** | *Frontend Developer* — *Cairo | June 2025 – Present*
  - **Portfolio Platform Development**: Developed comprehensive platform with event pages, admin dashboard for event creation, role-based access for instructors and students, location management, event galleries, and statistical analysis pages.
  - **HR Management & Judging System**: Built HR system managing team attendance with custom form builder and reviewer functionality similar to Google Forms, alongside event judging system for evaluating and assessing teams applying to events.
- **Physics Helping District (PhD)** | *Team Leader* — *Cairo | June 2022 – 2024*
  - **Mentorship**: Led peer tutoring initiative in electronics, achieving measurable academic improvement.
- **NASA Space Apps Challenge** | *Participant – Project: "Orrery"* — *Cairo | Oct 2024*
  - **Orrery App**: Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.

Projects

- **5-Stage Pipelined Processor (RISC)** | *VHDL, QuestaSim, Computer Architecture* — Source Code | *2025*
  - **Core Architecture**: Engineered a 32-bit 5-stage pipelined processor in VHDL, supporting a custom ISA of over 25 instructions.
  - **Hazard Resolution**: Optimized throughput by implementing Hazard Detection and Data Forwarding units to resolve control and data dependencies.
  - **Verification**: Validated functional correctness through rigorous simulation and testbench environments in QuestaSim.
- **CNN Convolution Accelerator** | *Verilog, OpenLane2, ASIC Design* — Source Code | *2025*
  - **Architecture**: Designed a synthesizable 2D convolution accelerator using Systolic Arrays and fully pipelined dataflow to maximize MAC throughput.
  - **Memory Hierarchy**: Implemented DMA-based loading and dual-port SRAM buffering to support up to 16x16 kernel execution.
  - **Physical Design**: Executed RTL-to-GDSII synthesis using OpenLane2, validating physical feasibility and achieving timing closure.
- **AES Encryption/Decryption Engine** | *Verilog, FPGA, Cryptography* — Source Code | *2023*
  - **Core Logic**: Synthesized robust AES cryptographic core in Verilog HDL supporting 128-bit, 192-bit, and 256-bit key standards.
  - **Datapath**: Engineered hardware-optimized modules for SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
  - **Validation**: Verified cryptographic correctness against standard NIST test vectors using rigorous debug tracing.

- **Tactic (Robotics & AI)** | *Embedded Systems* Source Code | *2024*
  - **System Integration**: Developed browser-controlled robotics suite integrating AI computer vision with low-latency hardware control.
  - **Firmware**: Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control and telemetry.
  - **Backend Processing**: Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.

- **Tactic (Robotics & AI)** | *Real-Time Systems, WebSockets, Embedded* Source Code | *2024*
  - **Real-Time Messaging System**: Engineered a low-latency WebSocket communication layer to integrate a web dashboard with embedded hardware.
  - **Reliability**: Ensured reliable message delivery and system synchronization between the browser interface and the robotic unit.

Technical Skills

- **Digital Design & Verification**: Verilog HDL, VHDL, RTL Design, RISC-V ISA, MIPS Architecture, Pipeline Design, RTL-to-GDSII Flow, OpenLane2, DRC/LVS Verification, Static Timing Analysis, FSM Design

- **Development Tools & Platforms**: ModelSim, QuestaSim, Intel Quartus Prime, GTKWave, ESP32, AVR/Arduino

- **Programming Languages**: C/C++ (Advanced), Python, x86 Assembly, MIPS Assembly